

COURSE : STIQK2113

DISCRETE STRUCTURE

5.1 FUNCTIONS
5.2 PIGEONHOLE PRINCIPLE
5.3 PERMUTATION FUNCTIONS

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5.1 FUNCTIONS

FUNCTIONS

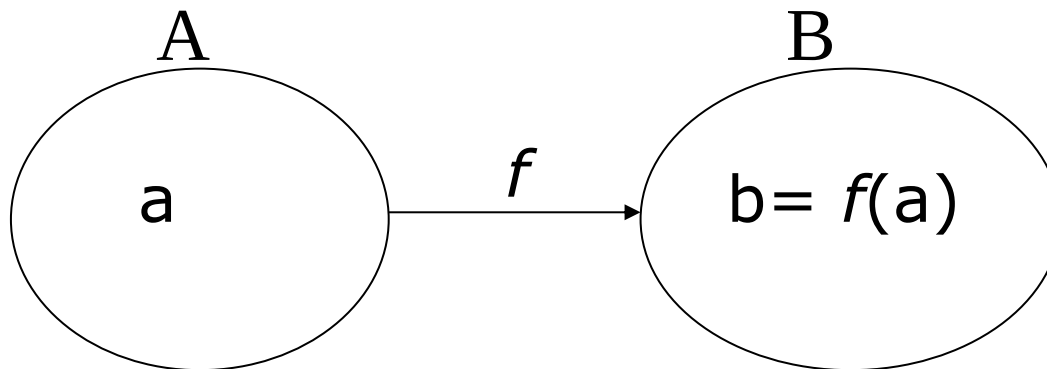
Let **A** and **B** be nonempty sets. A function **f** from **A** to **B** , is a relations from **A** to **B** such that for all **$a \in \text{Dom}(f)$** , **$f(a)$** , the f -relative set of **a** , contains just one element of **B** .

We write **$f(a) = b$** , where **$b \in B$**

The relation **f** can be described as the set of pairs **$\{(a, f(a)) \mid a \in \text{Dom}(f)\}$**

FUNCTIONS

- are also called mappings / transformation
- Can be viewed as rules that assign to each element $\mathbf{a} \in \mathbf{A}$ the unique element of $\mathbf{f(a)} \in \mathbf{B}$



- The element $\mathbf{a}$ is called an **argument** of the function $\mathbf{f}$, and $\mathbf{f(a)}$ is called the **value** of the function for the argument $\mathbf{a}$ and is referred to as the **image** of $\mathbf{a}$ under $\mathbf{f}$.

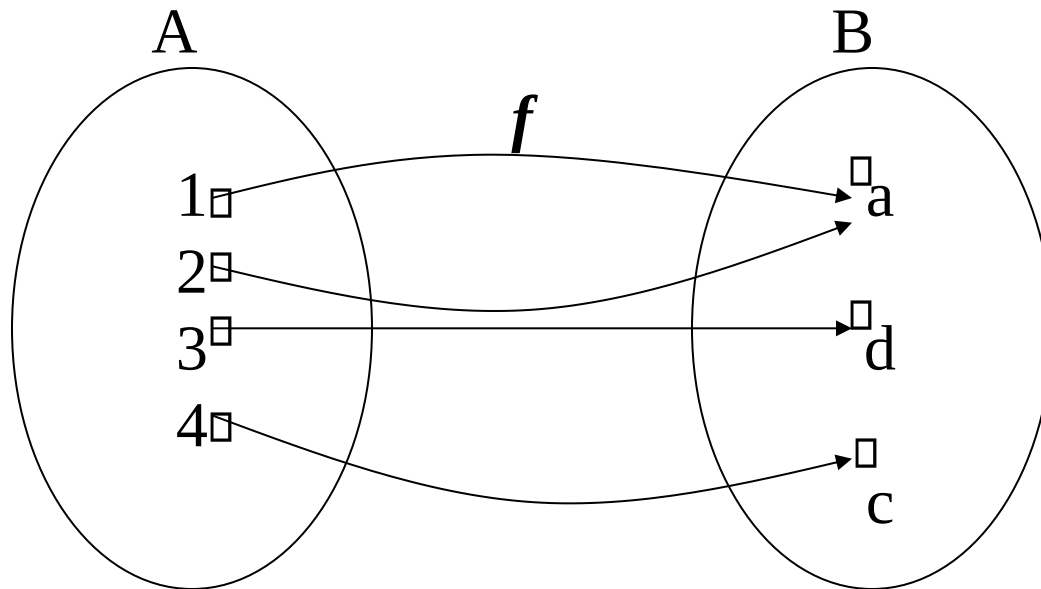
Example 1

Let $A = \{1,2,3,4\}$ and $B = \{a,b,c,d\}$, and

let $f = \{(1,a),(2,a),(3,d),(4,c)\}$

$\therefore$ we have

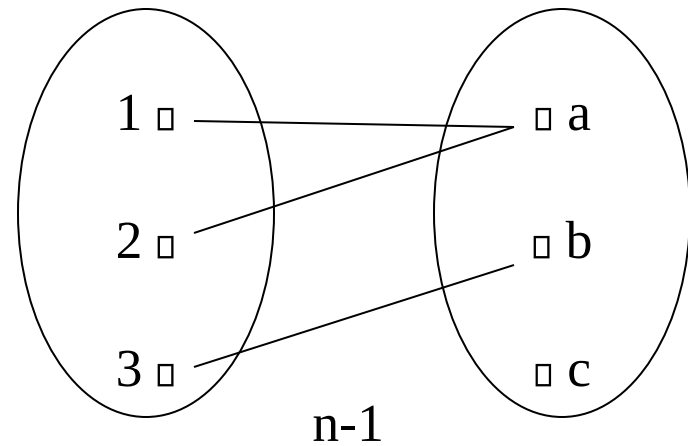
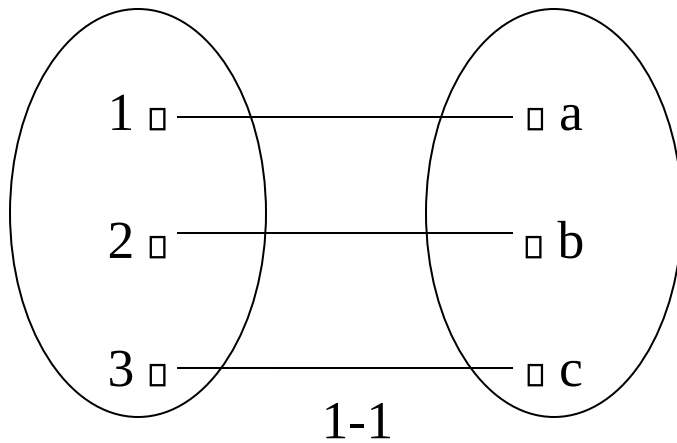
$f(1)=a, f(2)=a, f(3)=d, f(4)=c.$



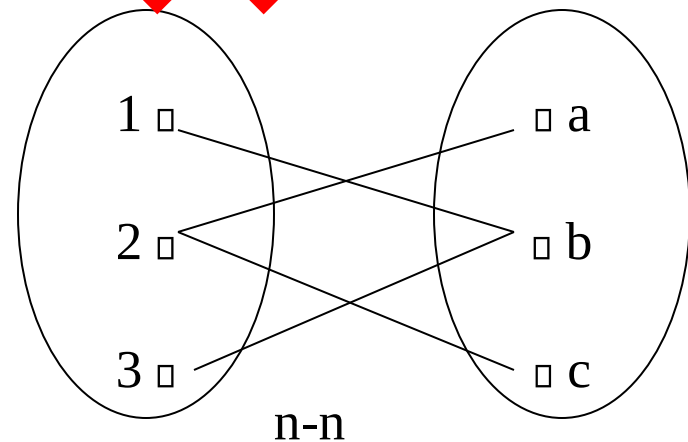
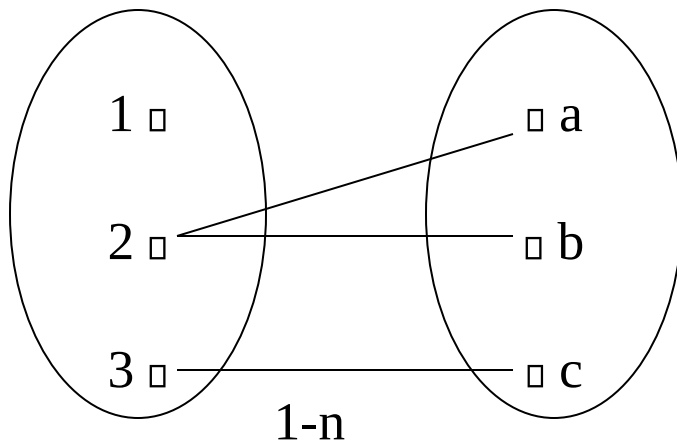
Since each set $f(n)$ is a single value, f is a function

4 types of relations

Functions



Not Functions



Example 2

Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$ consider the relations:
 $R = \{(1, x), (2, x)\}$ and $S = \{(1, x), (1, y), (2, z), (3, y)\}$.

Is R and S a function ?

Answer:

R is n-1 so R is a function

S is n-n so S is not a function

Composition of Functions

If $\mathbf{g}$ is a function from set A to set B and $\mathbf{f}$ is a function from set B to set C. The composition of functions $\mathbf{f}$ and $\mathbf{g}$ is denoted by $f \circ g$ and defined as

$$(f \circ g)(a) = f(g(a))$$

Example 3

Let $A = B = \mathbb{Z}$ and C is set of even numbers. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ defined by $f(a) = a + 1$ and $g(b) = 2b$ respectively.

Find **(a)** $f \circ g(a)$

Solution

$$= f(g(a))$$

$$= f(2a)$$

$$= 2a + 1$$

(b) $g \circ f(a)$

Solution

$$= g(f(a))$$

$$= g(a + 1)$$

$$= 2(a + 1)$$

$$= 2a + 2$$

Exercise 1

Let $A = B = C = \mathbb{R}$ Let $f : A \rightarrow B$ and $g : B \rightarrow C$ defined by $f(a) = a - 1$ and $g(b) = b^2 + 3$ respectively.

Find (a) $g \circ f(2)$

(b) $f \circ g(2)$

(c) $f \circ g(y)$

Exercise 2 (H.W)

Let $A = B = C = \mathbb{R}$ Let $f : A \rightarrow B$ and $g : B \rightarrow C$ defined by $f(x) = 4x + 1$ and $g(x) = 2x^2 - x + 3$ respectively.

Find **(a)** $g \circ f(x)$

(b) $f \circ g(x)$

SPECIAL TYPES OF FUNCTIONS

Let f be a function from A to B . Then we say that:

1. f is **everywhere defined** if $Dom(f) = A$.
2. f is **onto** if $Ran(f) = B$
3. f is **one to one** if we cannot have $f(a) = f(a')$ for two distinct elements a and a' of A

if $f(a) \neq f(a')$ if $a \neq a'$

and if $f(a) = f(a')$ if $a = a'$

Example 1

Let $f = \{(1, a), (2, a), (3, d), (4, c)\}$ where $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$.

Is f *everywhere* defined, *onto* or *one to one*?

Answer:

f is **everywhere defined** because

$$\text{Dom}(f) = \{1, 2, 3, 4\} = A.$$

f is **not onto** because $\text{Ran}(f) = \{a, c, d\} \neq B$.

f is **not one to one** because

$$f(1) = f(2) = a \quad \text{but } 1 \neq 2.$$

Example 2

Let $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2, b_3\}$, $C = \{c_1, c_2\}$ and $D = \{d_1, d_2, d_3, d_4\}$

Consider the following four functions, from A to B, A to D, B to C, and D to B. respectively.

- a) $f_1 = \{(a_1, b_2), (a_2, b_3), (a_3, b_1)\}$
- b) $f_2 = \{(a_1, d_2), (a_2, d_1), (a_3, d_4)\}$
- c) $f_3 = \{(b_1, c_2), (b_2, c_2), (b_3, c_1)\}$
- d) $f_4 = \{(d_1, b_1), (d_2, b_2), (d_3, b_1)\}$

Determine whether each function is one to one, onto or everywhere defined

Example 2 (Answer)

- a) f_1 is everywhere defined, one to one and onto.
- b) f_2 is everywhere defined and one to one, but not onto.
- c) f_3 is everywhere defined and onto but not one to one.
- d) f_4 is not everywhere defined, not one to one, and not onto.

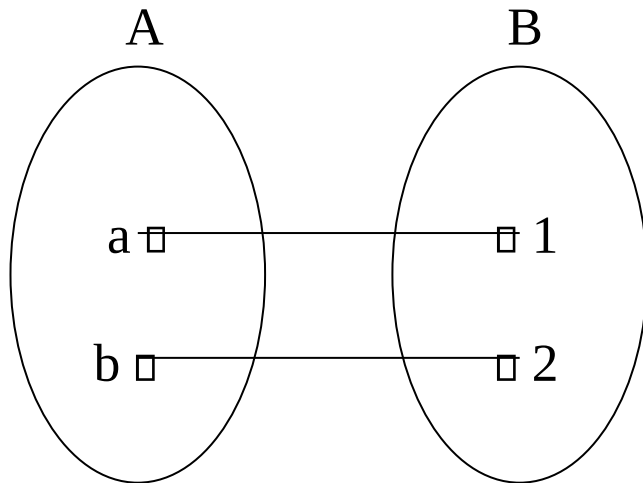
Bijection

If $f: A \rightarrow B$ is a one to one function. Then f assigns to each element a of $Dom(f)$ an element $b = f(a)$ of $Ran(f)$. Every b in $Ran(f)$ is matched, in this way, with one and only one element of $Dom(f)$.

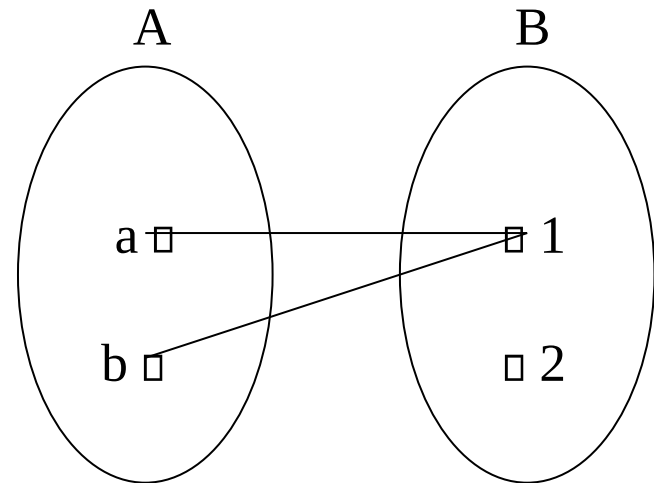
*For this reason, such an f is often called a **bijection** between $Dom(f)$ and $Ran(f)$.*

If f is also **everywhere defined and onto**, then f is called **one-to-one correspondence** between A and B .

Example 3



Bijection



Not Bijection

Invertible Functions

A function $f: A \rightarrow B$ is said to be invertible if its inverse relation f^{-1} , is also a function.

Example 4:

Let $A = \{1,2,3,4\}$ and $B = \{a,b,c,d\}$.

If $f_1 = \{(1,a), (2,a), (3,c), (4,d)\}$ and

$f_2 = \{(1,a), (2,c), (3,b), (4,d)\}$,

Is f_1 and f_2 invertible function ?

Example 4:

Let $A = \{1,2,3,4\}$ and $B = \{a,b,c,d\}$.
If $f_1 = \{(1,a), (2,a), (3,c), (4,d)\}$ and
 $f_2 = \{(1,a), (2,c), (3,b), (4,d)\}$,
Is f_1 and f_2 invertible function ?

Example 4 (Answer)

$$f_1^{-1} = \{(a,1), (a,2), (c,3), (d,4)\}$$

f_1^{-1} is not a function, since f_1^{-1} is 1-n where

$$f_1^{-1}(a) = \{1,2\},$$

$\therefore f_1$ is not an invertible function.

$$f_2^{-1} = \{(a,1), (c,2), (b,3), (d,4)\}$$

f_2^{-1} is a function, since f_2^{-1} is 1-1.

$\therefore f_2$ is an invertible function.

Example 5

Let $f: A \rightarrow B$ defined by :

(a) $f(a) = a + 1$

(b) $f(a) = a^3 + 1$

Find f^{-1}

Answer :

(a) $a + 1 = b$

$$a = b - 1$$

$$\therefore f^{-1}(b) = b - 1$$

(b) $a^3 + 1 = b$

$$a^3 = b - 1$$

$$a = \sqrt[3]{b - 1}$$

$$\therefore f^{-1}(b) = \sqrt[3]{b - 1}$$

Exercise 1

Let $f: A \rightarrow B$ defined by $f(x) = \frac{x+1}{2}$, find f^{-1}

Answer :

$$\frac{x+1}{2} = y$$

$$x+1 = 2y$$

$$x = 2y - 1$$

$$\therefore f^{-1}(y) = 2y - 1$$

5.2 PIGEONHOLE PRINCIPLE

THEOREM 1

Let $f: A \rightarrow B$ with a finite domain and range

Let:

$$|\text{Dom}(f)| = n \quad \text{and}$$

$$|\text{Ran}(f)| = m$$

Therefore :

a) If f is 1-1, then $m = n$.

b) If f is not 1-1, then $m < n$.

PIGEONHOLE PRINCIPLE

From theorem 1, note that $m < n$ is true when f is not 1-1.

Theorem 1 produces the following principle :

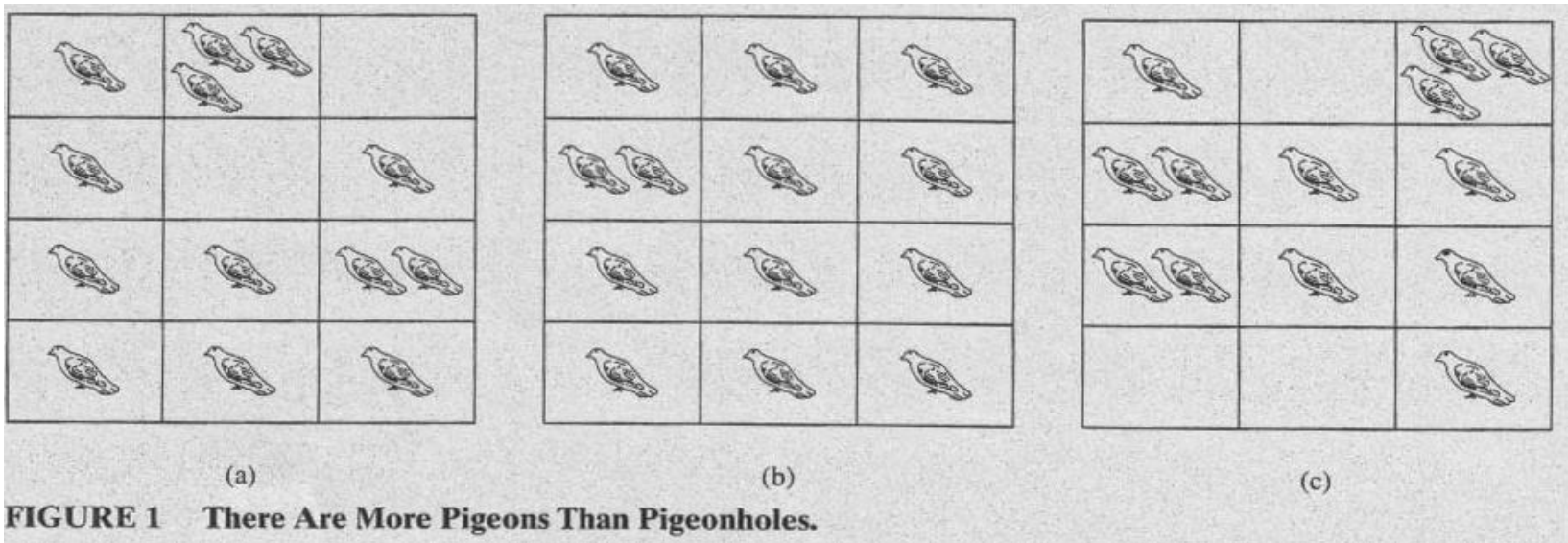
“ If $|\text{Ran}(f)| < |\text{Dom}(f)|$, then there will be 2 distinct element, a_1 and a_2 in $\text{Dom}(f)$ where $f(a_1) = f(a_2)$ ”.

This is the **PIGEONHOLE PRINCIPLE**.

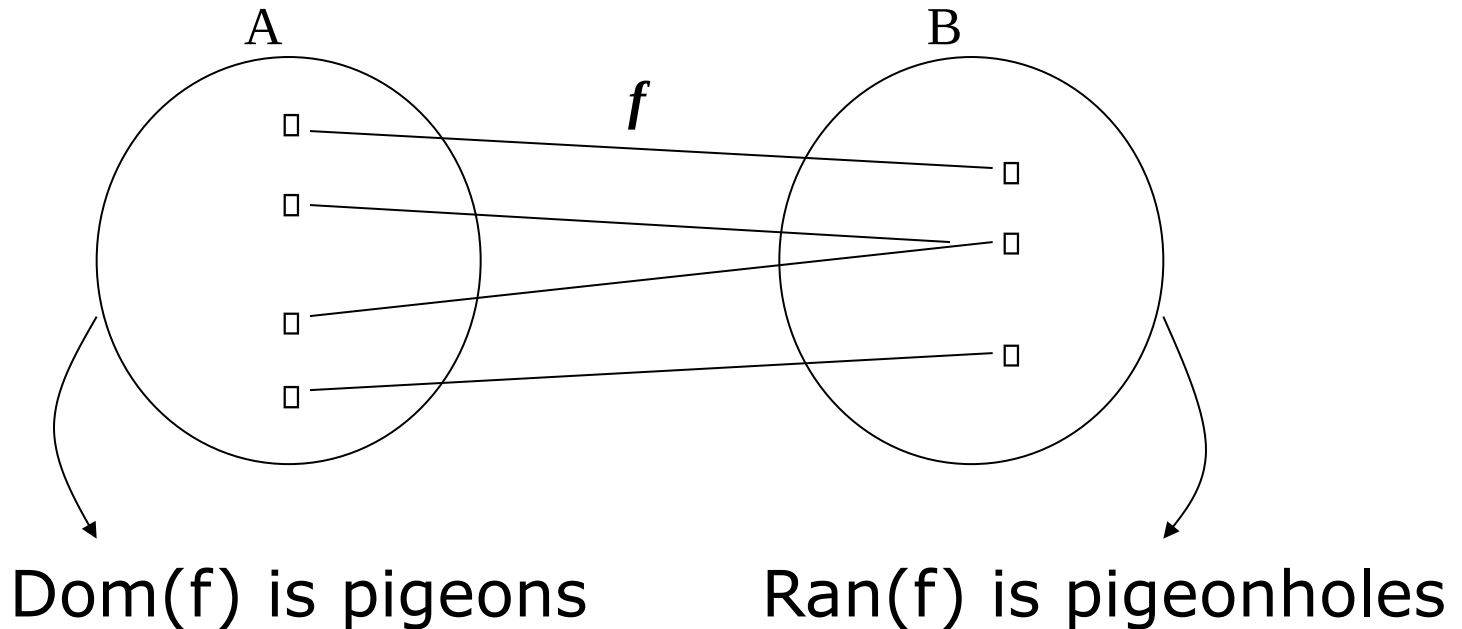
PIGEONHOLE PRINCIPLE

If there is more pigeons (n) than pigeonholes (m),
($m < n$),

Then, at least one pigeonholes must have two or more pigeons.



PIGEONHOLE PRINCIPLE



The function f is the rule that distribute each pigeons to a pigeonhole.

Example 1

From a group of 367 people, show that at least 2 people have the same birthday.

Answer :

- There are 366 days in a year, so 366 possible birthdays.
- 367 people are pigeons and 366 birth dates are pigeonholes.
- According to the pigeon hole principle, if pigeons are more than pigeonholes, then there will be at least 2 people with the same birthday.

Example 2

How many students are needed to ensure that **at least** 2 students get the same marks for an exam if the marking scale is 0 to 100?

Answer :

There are 0-100 marks, so 101 pigeonholes. To ensure at least 2 students get the same mark, there should be 102 students (pigeons).

The Extended Pigeonhole Principle

If n pigeons is assign to m pigeonhole,
then one of the pigeonhole must have at
least

$\lfloor (n-1)/m \rfloor + 1$ pigeons.

Note : $\lfloor 3/2 \rfloor = 1$

$\lfloor 9/4 \rfloor = 2$

$\lfloor 6/3 \rfloor = 2$

Example 1 : Extended PhP

From a group of 367 people, show that at least 2 people have the same birthday.

Answer :

Pigeonhole = $m = 366$ possible birthdays

Pigeons = $n = 367$ people

$$= \lfloor (n-1)/m \rfloor + 1$$

$$= \lfloor (367-1)/366 \rfloor + 1$$

$$= 1 + 1 = 2 \text{ pigeons in one pigeonhole}$$

$\therefore$ 2 people have the same birthday

Example 2 : Extended PhP

How many students are needed to ensure that at least 2 students get the same marks for an exam if the marking scale is 0 to 100?

Answer :

Pigeonhole = $m = 101$ possible marks

Pigeons = $n = ?$ people

$$= \lfloor (n-1)/m \rfloor + 1 = 2$$

$$= \lfloor (n-1)/101 \rfloor = 2-1$$

$$= n-1 = 101$$

$$= n = 102 \text{ pigeons}$$

$\therefore$ 102 students needed to ensure at least 2 students have the same marks

Example 3 : Extended PhP

Show that if any 5 numbers from 1 to 8 are chosen, then 2 of them at least will add to get 9.

Answer :

There are 4 different sets, each containing 2 numbers that add up to 9

$$A_1 = \{1,8\} \quad A_2 = \{2,7\} \quad A_3 = \{3,6\} \quad A_4 = \{4,5\}$$

Pigeonhole = $m = 4$ possible set of 2 numbers that add to 9.

Pigeons = $n = 5$ numbers chosen

$$= \lfloor (n-1)/m \rfloor + 1$$

$$= \lfloor (5-1)/4 \rfloor + 1$$

$$= 1 + 1 = 2 \text{ pigeons at least in one pigeonhole}$$

$\therefore$ 2 numbers will add up to 9.

Exercise 1

Using the extended pigeonhole principle, show that if there are 30 students in a class, then at least 2 of them have names starting with the same alphabet.

Exercise 2

How many students are needed in the TA2103 class to ensure that at least 6 students get the same grade if there are 5 possible grade (A,B,C,D and F)?

Answer:26

5.3 PERMUTATION FUNCTIONS

PERMUTATION FUNCTIONS

In this section we discuss bijections from a set **A** to itself.

A bijection from a set **A** to itself is called ***permutation of A*** .

Example 1

Let $A = \mathbb{R}$ and let $f: A \rightarrow A$ be defined by $f(a) = 2a + 1$.

Since f is **one to one** and **onto**, f is a permutation of A .

If $A = \{a_1, a_2, \dots, a_n\}$ is a finite set and p is a bijection on A , we list the elements of A and the corresponding function values $p(a_1), p(a_2), \dots, p(a_n)$ in the following form

$$\begin{pmatrix} a_1 & a_2 & \dots & a_n \\ p(a_1) & p(a_2) & \dots & p(a_n) \end{pmatrix} \rightarrow (1)$$

Example 1

Observe that (1) completely describes $\mathbf{p}$ since it gives the value of $\mathbf{p}$ for every element of $\mathbf{A}$. We often write

$$\mathbf{p} = \begin{pmatrix} a_1 & a_2 & \dots & a_n \\ p(a_1) & p(a_2) & \dots & p(a_n) \end{pmatrix}$$

Thus, if $\mathbf{p}$ is a permutation of a finite set $\mathbf{A} = \{a_1, a_2, \dots, a_n\}$, then the sequence $p(a_1), p(a_2), \dots, p(a_n)$ is just a rearrangement of the elements of $\mathbf{A}$.

Example 2

Let $A = \{1, 2, 3\}$. Then all the permutations of A are

$$1_A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \quad p_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \quad p_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$p_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad p_4 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad p_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

Example 3

Using the permutations of example 2, compute:-

(a) p_4^{-1}

(b) $P_3 \circ p_2$

Answer :

(a) viewing p_4 as a function

$$p_4 = \{(1,3), (2,1), (3,2)\}$$

$$p_4^{-1} = \{(3,1), (1,2), (2,3)\}$$

Or when written in increasing order of the first component of each ordered pair :-

$$p_4^{-1} = \{(1,2), (2,3), (3,1)\}$$

Thus

$$p_4^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = p_3$$

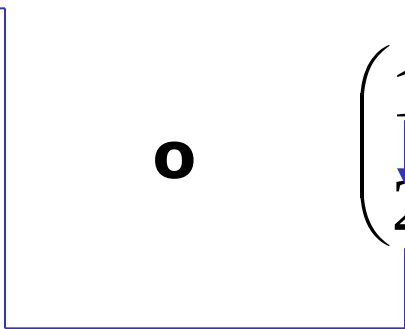
$$p_4 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

Example 3

$$p_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$p_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

(b) $p_3 \circ p_2$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$


$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = p_5$$

Exercise 1

If $A = \{1, 2, 3, 4, 5, 6\}$ and,

$$p_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 2 & 6 & 5 \end{pmatrix}$$

$$p_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 3 & 2 & 5 & 4 & 1 \end{pmatrix}$$

Find: **(a)** p_1^{-1}

(b) $p_2 \circ p_1$

(c) $p_1^{-1} \circ p_2^{-1}$

Theorem 1

If $\mathbf{A} = \{a_1, a_2, \dots, a_n\}$ is a set containing n elements, then there are $n!$ permutations of $\mathbf{A}$.

Let $b_1, b_2, \dots, b_r$ be r distinct elements of the set $\mathbf{A} = \{a_1, a_2, \dots, a_n\}$.

The permutation $\mathbf{p} : \mathbf{A} \rightarrow \mathbf{A}$ defined by

$$p(b_1) = b_2$$

$$p(b_2) = b_3$$

$$\vdots$$

$$p(b_{r-1}) = b_r$$

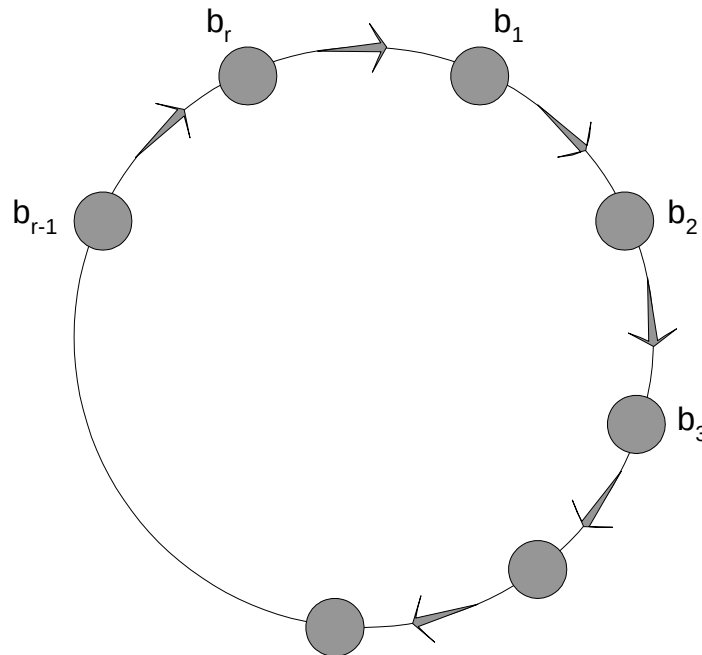
$$p(b_r) = b_1$$

$$p(x) = x, \text{ if } x \in \mathbf{A}, x \notin \{b_1, b_2, \dots, b_r\},$$

is called a cyclic permutation of length r and denoted by $(b_1, b_2, \dots, b_r)$.

Theorem 1

If the elements $b_1, b_2, \dots, b_r$ are arranged uniformly on a circle, as shown below, then a cycle p of length r moves these elements in a clockwise direction so that b_1 is sent to b_2 , b_2 to b_3 , ..., b_{r-1} to b_r and b_r to b_1 .



Example 4

- Let $A = \{1, 2, 3, 4, 5\}$. The cycle $(1, 3, 5)$ denotes the following permutation

$$P(1)=3$$

$$P(3)=5$$

$$P(5)=1$$

$$P(2)=2$$

$$P(4)=4$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix}$$

a cyclic permutation of length r denoted by $(b_1, b_2, \dots, b_r)$

$p : A \rightarrow A$ defined by

$$p(b_1) = b_2$$

$$p(b_2) = b_3$$

$$\vdots$$

$$p(b_{r-1}) = b_r$$

$$p(b_r) = b_1$$

$$p(x) = x,$$

if $x \in A, x \notin \{b_1, b_2, \dots, b_r\}$

- A cycle of length 1 $\Leftrightarrow$ Identity permutation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$

- Cycles are permutations, and like all permutations, they can be composed (combined via function composition).

Example 5

Let $A = \{1, 2, 3, 4, 5, 6\}$. Find:

(a) $(4, 1, 3, 5) \circ (5, 6, 3)$

(b) $(5, 6, 3) \circ (4, 1, 3, 5)$

Answer:

(a) Change the cycles to permutations

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 1 & 4 & 6 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 5 & 4 & 6 & 3 \end{pmatrix}$$
$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 4 & 1 & 6 & 5 \end{pmatrix}$$

Example 5

Answer :- Change the cycles to permutation

(b) $(5, 6, 3) \circ (4, 1, 3, 5)$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 5 & 4 & 6 & 3 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 1 & 4 & 6 \end{pmatrix} \\ = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 6 & 1 & 4 & 3 \end{pmatrix}$$

Observe that :

$$(4, 1, 3, 5) \circ (5, 6, 3) \neq (5, 6, 3) \circ (4, 1, 3, 5)$$

Example 6

Two cycles of a set A are said disjoint if no element of A appears in both cycles.

Let $A = \{1, 2, 3, 4, 5, 6\}$. Then:

- (a) the cycle $(1, 2, 5)$ and $(3, 4, 6)$ are disjoint, whereas
- (b) the cycle $(1, 2, 5) \circ (2, 4, 6)$ are not.

If $p_1 = (a_1, a_2, \dots, a_r)$ and $p_2 = (b_1, b_2, \dots, b_s)$ are disjoint cycles of A , then $p_1 \circ p_2 = p_2 \circ p_1$

Theorem 2

A permutation of a finite set that is not the identity or a cycle can be written as a product of disjoint cycles of length ≥ 2 .

Type	Two-line Notation	Cycle Notation	Notes
Identity	$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$	$()$ or omitted	No elements move
2-cycle	$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$	$(1\ 2)$	Swaps two elements
3-cycle	$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$	$(1\ 2\ 3)$	Rotates three elements
Single 5-cycle	$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 5 & 2 & 4 \end{pmatrix}$	$(1\ 3\ 5\ 4\ 2)$	One long cycle (length 5)

Example 7

Write the permutation $p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 6 & 5 & 2 & 1 & 8 & 7 \end{pmatrix}$ of the set $A = \{1,2,3,4,5,6,7,8\}$ as a product of disjoint cycles.

Answer:

- We start with 1 and find that $p(1) = 3$, $p(3) = 6$ and $p(6) = 1$ (**$1 \rightarrow 3 \rightarrow 6 \rightarrow 1$**), so we have cycle (1,3,6):
- Next, we choose the first element that is not in the first cycle : 2 and find that $p(2) = 4$, $p(4) = 5$, $p(5) = 2$ (**$2 \rightarrow 4 \rightarrow 5 \rightarrow 2$**), so, we have cycle (2,4,5)
- We now choose the first element that is not in the first and second cycle : 7 and find that $p(7) = 8$, $p(8) = 7$ (**$7 \rightarrow 8 \rightarrow 7$**). So, we have cycle (7,8)
- We can write p as a product of disjoint cycles as:

$$p = (7,8) \circ (2,4,5) \circ (1,3,6)$$

Even and Odd Permutations

A cycle of length 2 is called a transposition that is

$$p = (a_i, a_j)$$

Example : $(2,6), r = 2$

Every cycle can be written as a product of transpositions:

$$(b_1, b_2, \dots, b_r) = (b_1, b_r) \circ (b_1, b_{r-1}) \circ \dots \circ (b_1, b_3) \circ (b_1, b_2)$$

Example : $(1,2,3,4,5) = (1,5) \circ (1,4) \circ (1,3) \circ (1,2)$

(Even permutation)

Example 8

Write the permutation $p = (7,8) \circ (2,4,5) \circ (1,3,6)$ as a product of transposition.

Answer :

$$(1,3,6) = (1,6) \circ (1,3)$$

$$(2,4,5) = (2,5) \circ (2,4)$$

So,

$$p = (7,8) \circ (2,5) \circ (2,4) \circ (1,6) \circ (1,3)$$

(Odd permutation)

Exercise 2

Let $A = \{a,b,c,d,e,f,g\}$. Find the product of

(a) $(a,f,g) \circ (b,c,d,e)$

(b) $(f,g) \circ (b,c,f) \circ (a,b,c)$

And write the answer as:

1. a product of disjoint cycle
2. a product of transposition.

Exercise 3

Let $A = \{1,2,3,4,5\}$. Let $f = (5,2,3)$ and $g = (3,4,1)$ are the permutations of A .

Find

(a) $f \circ g$

(b) $f^{-1} \circ g^{-1}$

And write the answer as:

1. a product of disjoint cycle
2. a product of transposition.

Thank you